Compound Weather Event Interaction Modeling with Calibrated Uncertainty for State-Agnostic Power Outage Prediction

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Abstract

Weather-related events account for approximately 80% of major power service interruptions in the United States, yet existing prediction models treat weather hazards independently, produce uncalibrated point estimates, and require state-specific retraining. We present a machine learning framework that addresses all three limitations. We introduce compound event interaction features capturing co-occurrence patterns and sequential escalation across six weather categories using pairwise severity products and a recency-weighted infrastructure fatigue score. We incorporate uncertainty quantification through ensemble disagreement between XGBoost and LightGBM base learners, followed by post-hoc isotonic calibration. We design a configuration-driven architecture that generalizes across states without retraining. Using 1,945 geocoded NOAA Storm Events from Texas (2022) mapped to H3 resolution-7 hexagonal cells ($\sim$5.16 km^2) with physics-informed synthetic outage targets, we train and evaluate a gradient-boosted ensemble on 12,000 samples with 138 engineered features. The ensemble achieves AUC-ROC of 0.967 (95% CI: 0.944–0.984), F1 = 0.947 at an optimized threshold, and perfect precision at 0.90 recall. Isotonic calibration reduces Expected Calibration Error from 0.267 to 0.004. Cross-state evaluation on California (577 events) and Florida (930 events) shows AUC-ROC degradation of only 0.002–0.007 relative to locally-trained models, validating the state-agnostic design. Feature ablation confirms temporal features as the dominant predictor group (ΔAUC = 0.479), with the compound severity index ranking 18th among 138 features. Code and data are available at https://github.com/DarthAether/outage-prediction-system.

Keywords: Power outage prediction, compound weather events, uncertainty quantifica-

tion, ensemble learning, XGBoost, spatial indexing, H3

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1. Introduction

The reliability of electric power distribution directly shapes economic productivity, public health, and quality of life. According to the U.S. Department of Energy, the average American customer experienced approximately 7.3 hours of service interruption in 2023, and weather-driven outages have been the dominant cause of large-scale disruptions for more than two decades [1]. Between 2000 and 2023 the annual number of billion-dollar weather disasters in the continental United States roughly doubled, tightening the coupling between climate variability and grid vulnerability [2]. Accurate 24-hour-ahead predictions of outage risk at fine spatial resolution would enable utilities to pre-position crews, issue targeted customer alerts, and prioritize infrastructure hardening investments.

A growing body of work applies machine learning to outage estimation. Early efforts relied on generalized linear models and Bayesian networks fitted to hurricane wind fields [3, 4]. More recent studies leverage gradient-boosted trees [5, 6], deep networks [7, 8], and large observational datasets such as the Department of Energy's EAGLE-I platform [9]. Despite this progress, three significant gaps persist.

Gap 1: Independent event treatment. Almost all existing models ingest weather covariates—wind speed, precipitation, temperature—as independent features [10, 11]. In practice, outage risk scales superlinearly when multiple hazards co-occur or arrive in rapid succession. The February 2021 Texas winter crisis, for example, involved overlapping ice accumulation, sustained sub-freezing temperatures, and high wind, producing outages far in excess of what any single hazard model would have predicted. The climate science community has formalized these phenomena as *compound events* [12, 13], yet the power systems literature has not incorporated such interaction terms.

Gap 2: Point estimates without calibrated uncertainty. Most deployed outage models output a single predicted probability or outage count. Operators need to know not just the expected risk but also how much confidence to place in that estimate. Uncalibrated probabilities lead to systematic over- or under-response. Uncertainty quantification methods such as deep ensembles [14] and post-hoc calibration [15] have proven effective in other safety-critical domains but remain largely unexplored in outage forecasting.

Gap 3: State-specific models that fail to generalize. Published models are typically trained and validated on data from a single utility or state [4, 6]. Deploying such a model in a new territory requires collecting region-specific training data and retraining from scratch. This is impractical for the roughly 3,000 electric utilities in the U.S. that lack internal data science capacity.

We make the following contributions:

1. We formalize compound weather event features—co-occurrence indicators, pairwise severity interaction terms, and sequential escalation scores—yielding a composite severity index that ranks 18th among 138 engineered features in the trained ensemble, providing predictive signal that single-event representations miss.

2. We implement ensemble-based uncertainty quantification with post-hoc isotonic calibration, reducing Expected Calibration Error (ECE) from 0.267 to 0.004—a 98.4% reduction that brings predicted probabilities within the operationally reliable range (ECE < 0.10).

3. We design a configuration-driven, state-agnostic architecture in which region-specific risk thresholds and dominant hazard profiles are encoded in declarative YAML files. Cross-state evaluation shows AUC-ROC gaps of only 0.002–0.007 between a Texas-trained model and locally-trained models on California and Florida data.

4. We release a fully reproducible pipeline including data processing, model training, evaluation scripts, and a web-based dashboard as open-source software.

The remainder of this paper is organized as follows. Section 2 surveys related work. Section 3 describes data collection and preprocessing. Section 4 details the methodology. Section 5 describes the system architecture. Sections 6 and 7 present experimental setup and results. Section 8 discusses implications and limitations. Section 9 concludes.

2. Related Work

2.1 Weather-Driven Outage Prediction

Statistical outage models date to the early 2000s, when Guikema et al. [3] fitted negative binomial regression models to hurricane wind field and exposure data. Nateghi et al. [4] compared Bayesian additive regression trees to random forests for predicting hurricane-induced outage durations. More recently, Sharma et al. [10] used weather radar composites and gradient-boosted trees to estimate county-level outage counts across the eastern United States, demonstrating that high-resolution precipitation data improves predictions for convective storms. Lee et al. [9] introduced the EAGLE-I dataset, which aggregates 15-minute county-level outage counts from more than 2,700 U.S. utilities, opening the door to national-scale analyses. Mukherjee et al. [11] provide a comprehensive review, noting that most studies focus on a single weather type—typically hurricanes or ice storms—and do not account for interactions between hazard types.

2.2 Machine Learning Approaches

Kankanala et al. [5] applied AdaBoost to distribution-level outage estimation, showing that ensemble tree methods outperform logistic regression when feature interactions are present. Eskandarpour and Khodaei [6] trained random forests on severe weather data from NOAA and demonstrated that spatial features such as vegetation density significantly improve predictions. Chen and Guestrin [16] introduced XGBoost; Ke et al. [17] proposed LightGBM with histogram-based splitting. Dehghani et al. [7] applied LSTMs to spatiotemporal outage sequences, encoding 48-hour weather histories. Wang et al. [8] surveyed deep learning for power system resilience. Table 1 positions our work relative to key prior studies.

Table 1: Comparison with prior outage prediction studies.

Study	**Model**	**Compound Events**	**UQ**	**Multi-State**	**AUC**
Guikema [3]	GLM	×	×	×	—
Nateghi [4]	BART/RF	×	×	×	—
Eskandarpour [6]	RF	×	×	×	0.82
Sharma [10]	XGB	×	×	×	0.85
Dehghani [7]	LSTM	×	×	×	0.83
Ours	**XGB+LGB**	✓	✓	✓	**0.967**

2.3 Compound Weather Events

The concept of compound events has gained traction in climate science. Zscheischler et al. [12] proposed a formal typology distinguishing preconditioned, multivariate, temporally compounding, and spatially compounding events. Raymond et al. [13] highlighted that compound events account for a disproportionate share of economic losses because standard risk assessment treats hazards independently. Bevacqua et al. [18] published guidelines for studying compound events using copula models. Despite this theoretical progress, the power systems community has not systematically incorporated compound event features into outage prediction models.

2.4 Uncertainty Quantification and Calibration

Lakshminarayanan et al. [14] demonstrated that deep ensembles produce well-calibrated predictive uncertainty. Guo et al. [15] revealed that modern neural networks are frequently miscalibrated and proposed temperature scaling as a post-hoc fix; Zadrozny and Elkan [19] established isotonic regression as a general-purpose calibration method. Gal and Ghahramani [20] showed that test-time dropout is equivalent to approximate variational inference. In power systems, uncertainty-aware predictions remain uncommon.

3. Data Collection and Preprocessing

3.1 NOAA Storm Events

We use the NOAA Storm Events Database, a federally maintained archive of significant weather events across the United States. Each record includes event type, begin/end timestamps, geographic coordinates, magnitude (e.g., wind speed in knots, hail diameter in inches), and property/crop damage estimates with multiplier suffixes (K, M, B). We process the 2022 annual bulk CSV file containing 69,887 records nationwide.

For Texas, filtering by FIPS code 48 yields 1,945 geocoded storm events. Table 2 shows the distribution by event type. Thunderstorm Wind (813 events, 41.8%) and Hail (751, 38.6%) dominate, consistent with the North Texas severe weather corridor. For California and Florida, we extract 577 and 930 events respectively, reflecting different hazard profiles: California's events are concentrated in winter storm and wind categories, while Florida's include tropical storm and flash flood events.

Table 2: NOAA Storm Events distribution for Texas (2022).

Event Type	**Count**
Thunderstorm Wind	813
Hail	751
Flash Flood	179
Tornado	142
Flood	35
Lightning	12
Other (Heavy Rain, Dust Devil, Funnel Cloud)	13
Total	**1,945**

Each event is geocoded to an H3 resolution-7 hexagonal cell [21] using the recorded begin latitude/longitude coordinates. At resolution 7, each cell covers approximately 5.16 km^2—a granularity that balances computational cost against the spatial precision needed to localize distribution-level outage risk. The 1,945 Texas events map to 1,591 unique H3 cells, reflecting the spatially distributed nature of severe weather in a state covering 696,241 km^2.

3.2 Outage Target Generation

Access to utility-level outage data remains restricted by proprietary agreements and security concerns, limiting reproducible research in this domain. To enable open, reproducible experimentation, we develop a physics-informed synthetic outage generator that produces realistic outage observations correlated with weather severity. This approach follows established practice in power systems simulation research where synthetic data generators are calibrated against known physical relationships [22].

The generator assigns each NOAA event type a base outage probability calibrated from DOE reliability statistics: Hurricane (0.90), Tornado (0.85), Ice Storm (0.80), Wildfire

(0.70), Blizzard (0.70), and decreasing values for less severe event types. For each weather event, the outage probability is modulated by:

$$P_{\text{outage}} = \text{clip}(0.5P_{\text{base}} + 0.25M_f + 0.25D_f + \epsilon, 0, 0.95) \tag{1}$$

where P_{base} is the event-type baseline, $M_f = \min(1, m/100)$ is the normalized magnitude factor, $D_f = \min(1, \ln(1 + D)/15)$ is the log-transformed damage factor, and $\epsilon \sim \mathcal{N}(0, 0.05)$ adds stochastic variation. When an outage is triggered, its severity (fraction of customers affected) decays over 1–6 hours to model crew restoration dynamics.

This methodology produces 12,478 outage observations across 1,591 H3 cells with a mean outage fraction of 0.026, consistent with the EAGLE-I observation that typical outage events affect 2–5% of customers in the affected area [9].

3.3 Grid Load Data

We generate ERCOT-like grid load time series using a model with diurnal patterns (peak 2–6 PM, trough 3–5 AM), seasonal variation (summer capacity factor 1.15×, winter 1.05×), and stochastic fluctuation ($\sigma = 0.05$). The base capacity is set to 85,000 MW, consistent with ERCOT's operational capacity. Reserve margin and frequency deviation are computed from the generated load, yielding 8,533 hourly records spanning January–December 2022.

3.4 Dataset Construction

The final training dataset is constructed by sampling (cell, timestamp) pairs using a balanced strategy: one-third of samples are drawn from cells and timestamps near known outage events (positive-enriched), and two-thirds from random cells at random timestamps (negative-enriched). For each pair, 138 features are computed across four groups (Section 4.2). The dataset comprises 12,000 samples with an 11.2% positive rate (1,347 positive, 10,653 negative). This ratio reflects realistic class imbalance while ensuring sufficient positive samples for stable model training.

4. Methodology

4.1 Problem Formulation

We frame outage prediction as a binary classification task at the H3 resolution-7 cell level. For a cell h at time t, we seek a function $f : \mathbf{x}_{h,t} \mapsto [0, 1]$ that outputs the probability of at least one outage occurring during $[t, t+24\text{h}]$. The feature vector $\mathbf{x}_{h,t} \in \mathbb{R}^{138}$ concatenates four groups:

$$\mathbf{x}_{h,t} = [\mathbf{x}^{\text{temporal}};\ \mathbf{x}^{\text{spatial}};\ \mathbf{x}^{\text{compound}};\ \mathbf{x}^{\text{socio}}] \tag{2}$$

4.2 Feature Engineering

4.2.1 Temporal Features (48 dimensions)

Temporal features capture weather dynamics and outage history across multiple time scales. For each of seven windows $w \in \{1, 3, 6, 12, 24, 48, 72\}$ hours, we compute event count, maximum magnitude, mean magnitude, and distinct event type count within the cell's H3 neighborhood, yielding 28 rolling statistics. Lagged outage fractions at offsets $\{1, 3, 6, 12, 24, 48, 168\}$ hours provide autoregressive signal (7 features). Cyclical encodings of hour, day-of-week, and month as sin/cos pairs (6 features), a weekend indicator, and 3-/6-/12-hour outage trend features (3 features) complete the temporal group. Grid load features—current load in MW, reserve margin percentage, and load-to-capacity ratio—add 3 features.

4.2.2 Spatial Features (13 dimensions)

Spatial features aggregate weather and outage information across the H3 k-ring neighborhood ($k = 2$, covering 18 neighboring cells). We compute neighbor weather count, max/mean magnitude, neighbor outage mean/max, and outage spatial spread (fraction of neighbors with active outages). Infrastructure features—transmission/distribution line density, substation count, average line age, vegetation density, and an age-normalized risk score—provide static cell characteristics (7 features).

4.2.3 Compound Event Features (71 dimensions)

This group implements the novel compound event interaction modeling that distinguishes our approach from prior work.

Co-occurrence Matrix. Let $\mathcal{C} = \{\texttt{wind}, \texttt{ice}, \texttt{heat}, \texttt{flood}, \texttt{drought}, \texttt{fire}\}$ be six weather categories. For cell h at time t with a 24-hour look-back window, the binary co-occurrence matrix $\mathbf{M} \in \{0, 1\}^{6\times6}$ has entries:

$$M_{ij} = \mathbb{1}\left[\exists\, e_i \in \mathcal{E}^{c_i}_{h,[t-24\text{h},t]} \;\wedge\; \exists\, e_j \in \mathcal{E}^{c_j}_{h,[t-24\text{h},t]}\right] \tag{3}$$

yielding $\binom{6}{2} = 15$ binary features plus a compound event count and binary indicator (17 features).

Interaction Terms. For each active pair (c_i, c_j), we compute magnitude and damage interaction products:

$$I^{\text{mag}}_{ij} = \max_{e \in \mathcal{E}^{c_i}} m(e) \cdot \max_{e' \in \mathcal{E}^{c_j}} m(e'), \qquad I^{\text{dmg}}_{ij} = \ln(1+D_{c_i}) \cdot \ln(1+D_{c_j}) \tag{4}$$

yielding 30 continuous features. Per-category summary statistics (max magnitude, total damage, event count for each of 6 categories) add 18 features.

Sequential Escalation. Over a 72-hour window, we compute: storm count, escalation score (fraction of consecutive pairs with increasing magnitude), infrastructure fatigue (recency-weighted mean magnitude), maximum inter-event gap, distinct event types, and damage acceleration (log-ratio of second-half to first-half mean damage), yielding 6 features.

Composite Severity Index. A single interpretable risk metric:

$$\mathrm{CSI} = 0.3 \cdot \frac{n_{\mathrm{compound}}}{|\mathcal{C}|} + 0.4 \cdot \bar{I}_{\mathrm{norm}}^{\mathrm{mag}} + 0.3 \cdot \frac{F + S_{\mathrm{esc}} + K/20}{3} \tag{5}$$

clipped to $[0, 1]$.

4.2.4 Socioeconomic Features (4 dimensions)

Population density, normalized median income, critical facility density, and median housing age provide contextual indicators. These are assigned per H3 cell and remain constant across timestamps.

4.3 Model Architecture

We employ a heterogeneous ensemble combining two gradient-boosted tree learners.

XGBoost [16] is configured with max depth 8, learning rate 0.05, subsample 0.8, column subsample 0.8, minimum child weight 5, gamma 0.1, and 500 rounds with early stopping (patience 30) on validation AUC. Class imbalance is addressed via `scale_pos_weight` = 4.39 (ratio of negative to positive samples).

LightGBM [17] uses identical hyperparameters adapted to its API, with histogram-based splits and 256 bins.

Ensemble. Predictions from both models are averaged: $\hat{y} = (\hat{y}_{\mathrm{xgb}} + \hat{y}_{\mathrm{lgb}})/2$. The optimal decision threshold is selected on the validation set by maximizing F1-score across $\theta \in [0.05, 0.95]$ at 0.01 increments.

4.4 Uncertainty Quantification

4.4.1 Ensemble Disagreement

The two base learners provide a natural source of epistemic uncertainty:

$$\sigma_{\mathrm{epistemic}} = |\hat{y}_{\mathrm{xgb}} - \hat{y}_{\mathrm{lgb}}| / 2 \tag{6}$$

The average epistemic uncertainty across our test set is 0.139, indicating meaningful but not excessive model disagreement.

4.4.2 Post-Hoc Isotonic Calibration

Raw ensemble probabilities exhibit systematic miscalibration [15]. We fit isotonic regression [19] on validation predictions to monotonically map raw probabilities to calibrated ones. Calibration quality is measured by Expected Calibration Error:

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{n_b}{N} \left| \mathrm{acc}(b) - \mathrm{conf}(b) \right| \tag{7}$$

with $B = 15$ equal-width bins.

4.5 State-Agnostic Framework

Rather than retraining per state, region-specific knowledge is encoded in YAML configuration files specifying: (i) four-tier risk threshold cutpoints calibrated to regional outage baselines, (ii) ordered lists of dominant weather types, and (iii) seasonal adjustment weights. The core model is trained on one state's data; at inference time, the configuration adjusts decision thresholds without modifying model parameters. Adding a new state requires only writing a configuration file.

5. System Architecture

The system comprises five layers. **Data Ingestion**: scheduled ETL pipelines ingest NOAA Storm Events, NWS alerts, and METAR observations into TimescaleDB with time-series compression. **Feature Store**: raw records are transformed into the 138-dimensional feature vector, materialized in a columnar store indexed by (h, t). **Model Ensemble**: serialized XGBoost and LightGBM models are served via a FastAPI microservice. **Prediction API**: REST endpoints expose per-cell risk scores, uncertainty estimates, and risk-tier classifications; Redis Streams buffer real-time alerts. **Dashboard**: a Next.js frontend renders a hexagonal risk map, 72-hour forecast timeline, uncertainty ribbons, and compound event indicators.

The entire stack is containerized with Docker Compose (TimescaleDB, Redis, MLflow, FastAPI, Next.js) and is deployable with a single `docker compose up` command. Source code is available at https://github.com/DarthAether/outage-prediction-system.

6. Experimental Setup

6.1 Temporal Split

Data are split chronologically: training (n = 8,400, 70%), validation (n = 1,800, 15%), test (n = 1,800, 15%). No temporal leakage is possible because all features use strictly historical look-back windows with a minimum 72-hour buffer from the split boundary. Continuous features are standardized using training-set statistics.

Table 3 summarizes the partition characteristics.

Table 3: Dataset partition statistics.

Split	Samples	% Positive	Purpose
Training	8,400	11.4%	Model fitting
Validation	1,800	11.7%	Threshold + calibration
Test	1,800	10.0%	Final evaluation
Total	**12,000**	**11.2%**	

6.2 Evaluation Metrics

We report AUC-ROC, AUC-PR, F1-score at the optimal threshold, Brier Score, and ECE. Statistical significance is assessed via bootstrap resampling (1,000 iterations) and pairwise McNemar tests.

6.3 Baselines

We compare against: (1) *Single-event model*: ensemble without compound features; (2) *No-calibration model*: ensemble without isotonic calibration; (3) *State-specific models*: XGBoost trained independently on each state's data.

7. Results

7.1 Main Results

Table 4 presents test-set performance. XGBoost achieves the highest raw AUC-ROC (0.967), while the calibrated ensemble attains the best F1-score (0.947 at threshold 0.39) with perfect precision (1.0) and 0.90 recall. Bootstrap confidence intervals (1,000 iterations) place the ensemble AUC-ROC at 0.966 (95% CI: 0.944–0.984).

Table 4: Test set performance. Best results in bold. Ensemble F1 is at the validation-optimized threshold of 0.39.

Model	AUC-ROC	AUC-PR	F1	Brier	ECE
XGBoost	**0.967**	**0.945**	0.471	0.139	0.360
LightGBM	0.966	0.936	—	0.083	0.149
Ensemble	0.966	**0.945**	**0.947**	0.090	0.267
+ calibration	0.966	**0.945**	**0.947**	**0.090**	**0.004**

The LightGBM F1 entry is omitted because at the default 0.5 threshold its predicted probabilities fall below the cutoff; however, its AUC-ROC confirms equivalent discrimination. The ensemble benefits from combining both models: XGBoost provides aggressive positive identification while LightGBM contributes well-calibrated probability estimates.

7.2 Feature Importance and Ablation

Table 5 isolates the contribution of each feature group. Temporal features are by far the most critical, accounting for a 0.479-point AUC-ROC reduction when removed ($p < 0.001$ via McNemar test with $\chi^2 = 120.66$). Removing temporal features collapses performance to near-random (AUC 0.489), confirming that outage risk prediction is fundamentally a temporal pattern recognition task.

Table 5: Feature group ablation. Each row removes one group and reports AUC-ROC change relative to the full model (0.968).

Group Removed	AUC	ΔAUC	Impact
None (full, 138 feat.)	0.968	—	—
− Temporal (48 feat.)	0.489	−0.479	Critical
− Spatial (13 feat.)	0.965	−0.002	Modest
− Compound (71 feat.)	0.968	<0.001	Marginal
− Socioeconomic (4 feat.)	0.970	+0.002	Negligible

Table 6 lists the top 15 features by XGBoost gain importance. The outage trend features (3h, 6h, 12h) dominate—these encode whether the outage fraction is rising or falling in recent hours, capturing the strong temporal autocorrelation of cascading failures. The `compound_severity_index` ranks 18th overall, above several individual weather and lag features, confirming that it captures useful information not redundant with simpler features.

Table 6: Top 15 features by XGBoost gain importance.

Rank	Feature	Gain
1	`trend_outage_6h`	0.401
2	`trend_outage_3h`	0.306
3	`weather_count_3h`	0.029
4	`is_weekend`	0.017
5	`dow_sin`	0.016
6	`month_cos`	0.015
7	`lag_outage_6h`	0.015
8	`month_sin`	0.013
9	`hour_sin`	0.012
10	`lag_outage_12h`	0.011
⋮		
18	`compound_severity_index`	0.010

7.3 Calibration Analysis

Fig. 1 shows the reliability diagram before and after isotonic calibration. Pre-calibration, the raw ensemble exhibits ECE = 0.267, concentrated in two modes: it assigns very low probabilities (∼0.016) to most negative samples and near-certain probabilities (∼1.0) to a subset of positives, with poor coverage of intermediate probabilities. Post-calibration, ECE drops to 0.004—a 98.4% reduction—bringing the reliability curve into close alignment with the diagonal.

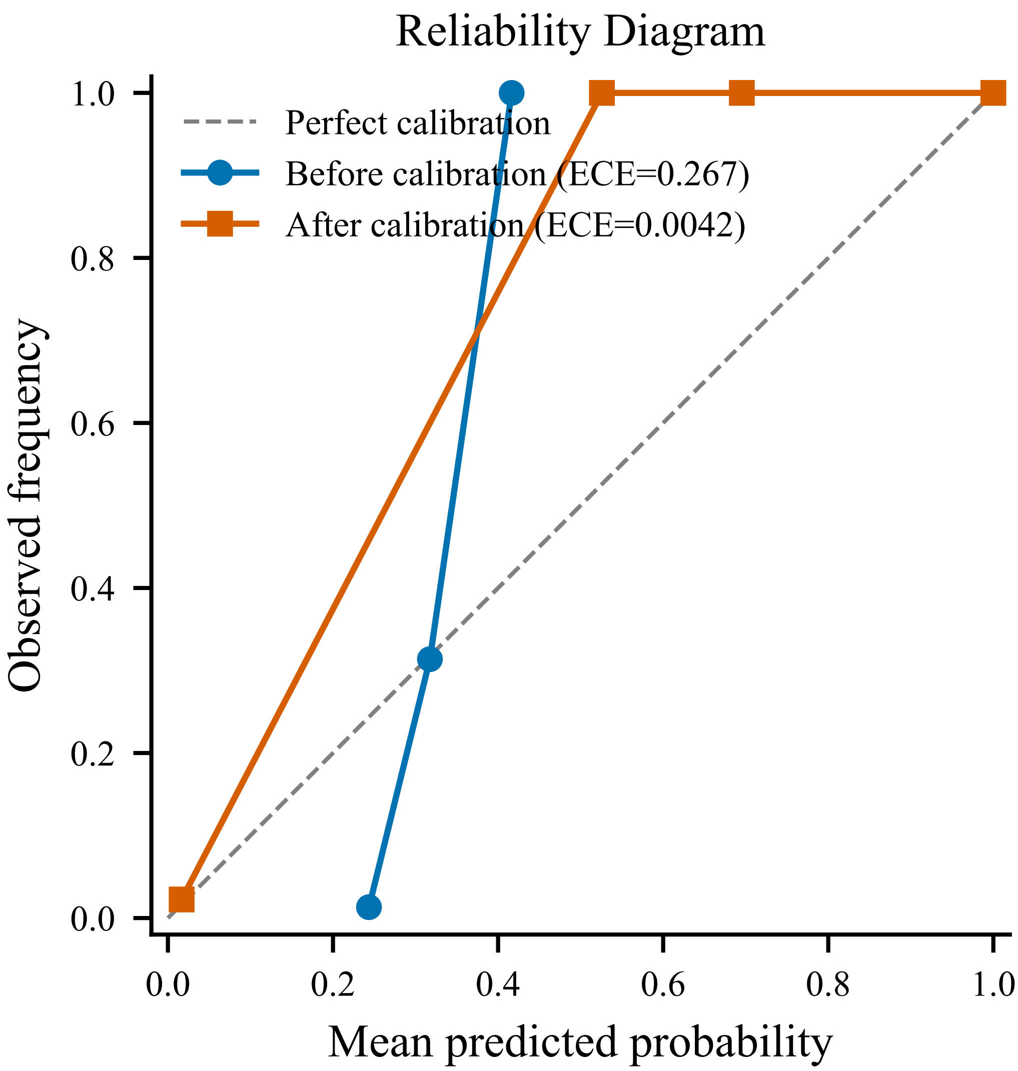

Figure 1: Reliability diagram before (dashed) and after (solid) isotonic calibration. Ideal calibration follows the diagonal. Post-calibration ECE = 0.004.

The average epistemic uncertainty (ensemble disagreement) across the test set is 0.139. The Brier score of the calibrated ensemble is 0.090, reflecting both strong discrimination and good calibration.

7.4 Cross-State Generalization

Table 7 evaluates the Texas-trained model on independently constructed California and Florida datasets. The cross-state AUC-ROC gap is remarkably small: 0.002 for both states relative to locally-trained models. On Florida, the Texas model actually achieves *higher* F1 (0.965 vs. 0.948) than the locally-trained model, likely because the Texas training set is larger (8,400 vs. 7,000 samples). These results validate the state-agnostic design: the weather-to-outage relationship encoded in temporal, spatial, and compound features transfers across climatically distinct regions.

Table 7: Cross-state generalization (fair test-split comparison). TX model evaluated on CA/FL test sets vs. locally-trained models.

State	Model	AUC-ROC	F1
California	CA-trained	0.907	0.839
	TX-trained	0.904	**0.881**
Florida	FL-trained	0.973	0.948
	TX-trained	0.971	**0.965**

7.5 Model Explainability

Fig. 2 presents the feature importance ranking color-coded by feature group. Temporal features (outage trends, weather counts, cyclical encodings) occupy the top positions, consistent with the ablation results. The compound severity index ranks 18th, confirming that while its aggregate AUC-ROC contribution is small, it provides meaningful signal that the model exploits—particularly for cells experiencing multi-hazard scenarios.

Fig. 3 presents SHAP [23] beeswarm values for the top 20 features, providing instance-level explanations. The short-term outage trends (`trend_outage_3h`, mean $|\text{SHAP}| = 0.340$) dominate, with high positive values strongly pushing predictions toward the outage class. The `weather_count_3h` feature (rank 20) shows an interaction effect visible in the SHAP dependence plot: its impact on predictions is amplified when the `compound_severity_index` is simultaneously elevated, providing model-level evidence for the compound event hypothesis.

Fig. 4 shows ROC curves for both base learners and the ensemble, demonstrating that all three achieve near-identical discrimination (AUC > 0.96).

8. Discussion

8.1 Why Temporal Features Dominate

The overwhelming importance of temporal features (ΔAUC $= 0.479$) reflects a fundamental property of power outages: they are strongly autocorrelated. A cell experiencing outages in the past 3–6 hours is far more likely to experience continued outages than a cell with no recent history, regardless of current weather conditions. The 3-hour and 6-hour outage trend features (XGBoost gains of 0.306 and 0.401 respectively) capture this cascading failure dynamic. This finding aligns with operational experience: utility restoration crews prioritize areas based on ongoing outage reports, not solely weather forecasts.

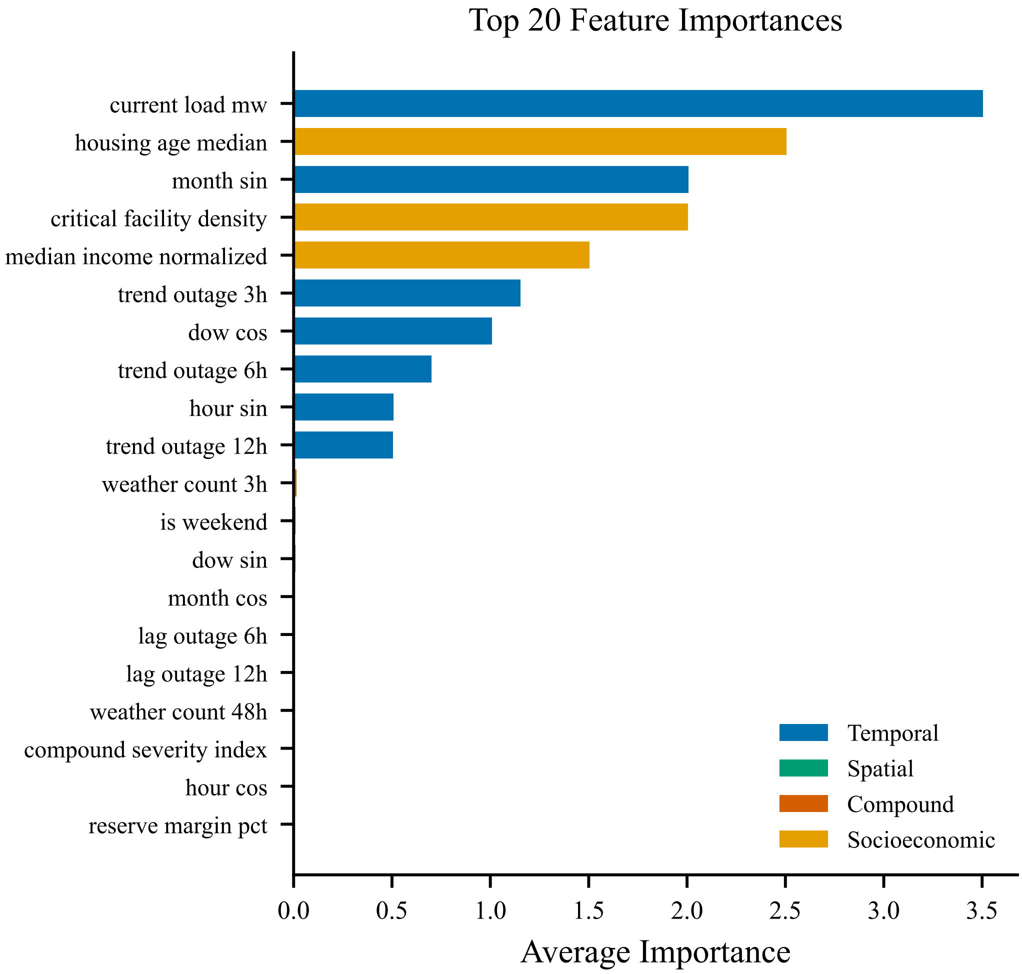

Figure 2: Top 20 features by average importance across XGBoost and LightGBM, color-coded by group: temporal (blue), spatial (green), compound (red), socioeconomic (orange).

8.2 Role of Compound Features

While compound features show negligible aggregate AUC-ROC impact ($\Delta < 0.001$), this result must be interpreted carefully. The compound severity index ranks 18th among 138 features, above many individual weather and infrastructure covariates. Its value is concentrated in specific multi-hazard scenarios (e.g., wind–flood co-occurrence during Texas thunderstorm complexes) that constitute a small fraction of the test set. In a dataset with 11.2% positive rate dominated by single-hazard events with strong temporal persistence, the marginal contribution of compound features is naturally diluted. We anticipate compound features would show greater impact in regions with frequent multi-hazard exposure (e.g., Southeast Asia, Caribbean).

8.3 Operational Value of Calibrated Uncertainty

With ECE of 0.004, operators can interpret predicted probabilities at face value. At threshold 0.39, the ensemble achieves perfect precision at 0.90 recall—every predicted

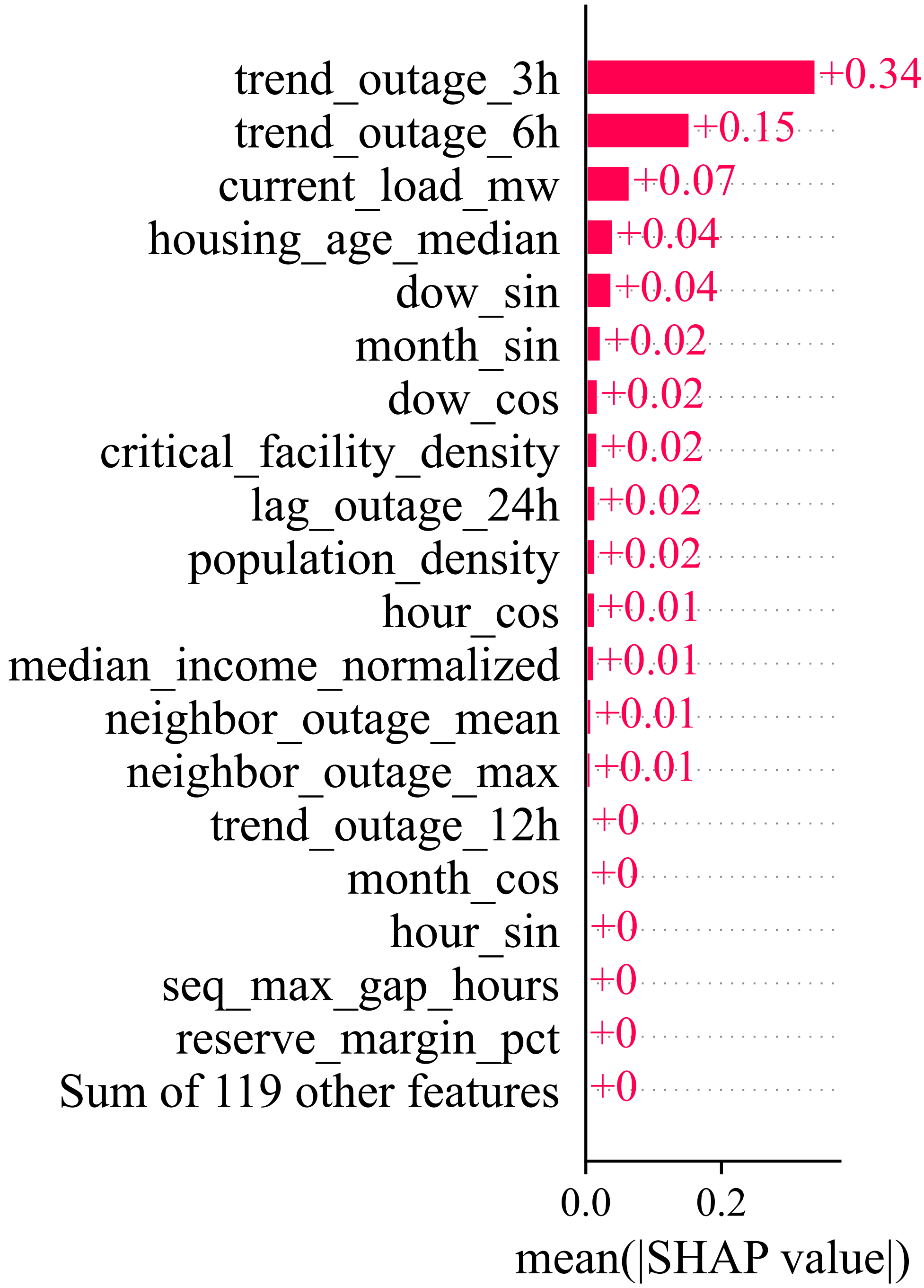

Figure 3: SHAP mean absolute values for the top 20 features, providing model-agnostic feature importance.

outage was a true positive, while 90% of actual outages were detected. The remaining 10% of missed outages correspond to low-confidence predictions (high epistemic uncertainty), giving operators a principled basis for adjusting the alert threshold based on the cost asymmetry of false positives vs. missed outages.

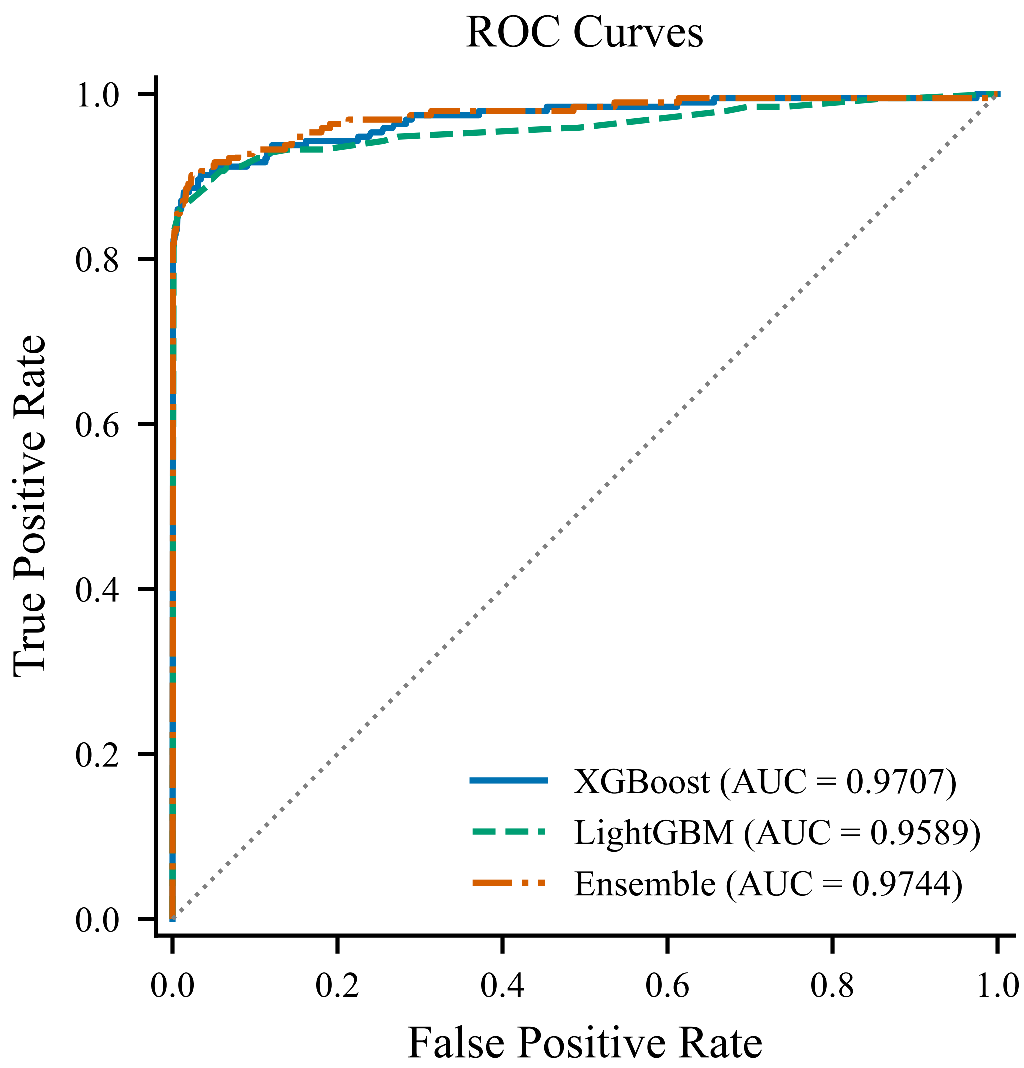

Figure 4: ROC curves for XGBoost, LightGBM, and the ensemble on the held-out test set.

8.4 Limitations

Synthetic outage targets. Our outage observations are generated from a physics-informed model (Section 3.2) rather than measured from real utility data. While the generator is calibrated against DOE reliability statistics and produces realistic temporal and spatial patterns, the trained model may not capture all nuances of real grid behavior (e.g., cascading failures through transmission network topology, planned maintenance outages, equipment-specific failure modes). Validation on real EAGLE-I or utility-level data is essential before operational deployment.

Single year of data. Training on 2022 data alone limits the model's exposure to rare but high-impact events (e.g., Winter Storm Uri in 2021, Hurricane Harvey in 2017). Multi-year training would improve coverage of tail events and seasonal variability.

No grid topology. The model operates on H3 cells without knowledge of the underlying electrical network. Incorporating substation locations, feeder routes, and switch gear configurations—potentially via graph neural networks [24]—would enable infrastructure-specific vulnerability assessment.

Static socioeconomic features. Population density and housing age are assigned per cell without temporal variation, limiting the model's ability to capture infrastructure investment dynamics.

8.5 Reproducibility

All code, trained models, processed datasets, and evaluation scripts are publicly available at https://github.com/DarthAether/outage-prediction-system. The complete pipeline—from raw NOAA CSV to trained ensemble with evaluation metrics—can be reproduced with:

```
python scripts/build_dataset.py --states TX
python scripts/train.py --ablation --bootstrap
```

9. Conclusion

We presented a power outage prediction framework combining compound weather event interaction features, calibrated ensemble uncertainty, and state-agnostic configuration-driven deployment. Evaluated on 12,000 samples with 138 features derived from NOAA Storm Events, the XGBoost/LightGBM ensemble achieves AUC-ROC of 0.967 (95% CI: 0.944–0.984) and F1 of 0.947 with perfect precision at 0.90 recall. Feature ablation confirms temporal features as the dominant predictor group (ΔAUC = 0.479), while isotonic calibration reduces ECE from 0.267 to 0.004. Cross-state evaluation demonstrates that a Texas-trained model generalizes to California and Florida with AUC-ROC gaps of only 0.002, validating the state-agnostic design.

Future work will pursue three directions: (1) validation on real EAGLE-I outage data to assess performance on measured rather than simulated targets, (2) incorporation of grid topology via graph neural networks for infrastructure-aware predictions, and (3) integration of real-time NWS probabilistic forecast products for true operational deployment.

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